Project Roadmap – Future Improvements

1. Introduction

The current release provides a working two-dimensional Brusselator reaction-diffusion solver with three numerical methods: Explicit Euler, RK4 and Implicit Backward Euler.

This roadmap outlines possible future improvements, known limitations and features deliberately excluded from the first release.

2. Planned Improvements

1. PVD-based time-dependent visualisation

Currently, the solver generates individual .vti files. A future improvement would be to generate a .pvd file that collects these time steps.

This would allow users to open a single file in ParaView and play the simulation as an animation.

2. Code improvements

- Expand unit and component test coverage.

3. Known Limitations

The current release has the following limitations:

- The solver is designed exclusively for the two-dimensional Brusselator reaction-diffusion equation.
- It does not support arbitrary reaction-diffusion equations.
- Time-dependent simulation results cannot currently be loaded and played directly as a movie in ParaView using a PVD collection file.
- Users must manually specify the grid resolution and time-step size; adaptive mesh refinement and adaptive time stepping are not implemented.

4. Features Deliberately Excluded from the First Release

Some features were intentionally left out to keep the initial implementation focused and manageable.

These include:

- General-purpose PDE solver capabilities.
- Advanced spatial discretisation methods.
- Adaptive mesh refinement.

- Adaptive time stepping.
- Parallel computing support.
- PVD-based time-series output.

These decisions were made to prioritise the implementation, testing and validation of the existing numerical methods.

5. Future Development Priorities

Priority	Improvement
High	PVD-based time-series output for ParaView
High	Investigate more accurate spatial and temporal discretisation methods
Medium	Adaptive time stepping

6. Conclusion

The primary objective of future development is to improve the solver's numerical accuracy, computational efficiency and usability.

The immediate priorities are to investigate improved discretisation methods and implement PVD-based output for time-dependent visualisation in ParaView.